

DM887 Reinforcement Learning: Assignment 1

River Swim med Online Model-Based RL

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9. marts 2026

Yes, my model can solve the task. As shown in the plot, all river lengths have been simulated five times, and the average reward is plotted. It is evident that the agent learns to reach the final state to collect the large reward. We can observe this as the learning curve flattens, indicating that the model has converged to the optimal policy and ceased further exploration.

I have implemented the R-MAX algorithm, which utilizes the Optimism in the Face of Uncertainty (OFU) principle. The model begins with an optimistic view where all unknown state-action pairs are assumed to yield the maximum possible reward.

I am not employing an ϵ -greedy approach. Instead, the R-MAX model balances exploration and exploitation by treating unknown states as highly rewarding, regardless of their true reward. I have set the threshold for a state-action pair to be considered 'known' to $m = 50$. Following the PAC (Probably Approximately Correct) analysis of model error, the number of samples m required to bound the error by ϵ with a probability of at least $1 - \delta$ follows $m \geq \frac{R_{max}^2}{2\epsilon^2} \log(K/\delta)$. Consequently, a larger m reduces the probability of approximation errors but significantly increases the required runtime.

Once a state-action pair has been visited m times, the model switches from the optimistic R_{max} assumption to using true empirical estimates, $\hat{P}$ and $\hat{r}$, to calculate the optimal policy.

I utilize Policy Iteration (PI), which is often preferable to Value Iteration because it converges in a finite number of steps due to the finite number of stationary policies. The discount factor γ is set to 0.999, ensuring convergence in this non-episodic environment. This effectively provides the agent with a long-term memory ('effective horizon').

I have implemented early stopping that triggers when all state-action pairs are known and the agent has reached the terminal state s_T , as no further learning can occur after this point. For the various river lengths ($T = 5, 10, 15, 20$), the model reaches this convergence at approximately 1000, 2000, 3000, and 4000 steps, respectively. This suggests that the empirical convergence time scales linearly with T . This performance is significantly better than the theoretical worst-case sample complexity upper bound of $\tilde{O}(|S|^2|A|T^4)$.

1 Appendix

